

Frequency-Adaptive Audio Limiting for Reduced Acoustic Exposure and Loudness Preservation

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Abstract—Conventional personal-audio limiters reduce listening level by applying one gain to every frequency. A-weighted exposure and programme loudness do not weight the spectrum in the same way, so it is reasonable to ask whether cutting some bands more than others can lower an exposure metric at the same loudness. This paper tests that question with a three-band digital processor: a complementary Linkwitz–Riley split at 300 Hz and 4 kHz, a low-band shelf of -12 dB with harmonic fill from 80 to 400 Hz, an A-weighted mid-band compressor, and a mild high-band envelope gain. On 20 clips (16 Freesound recordings and four synthetic signals) we compare the processor with frequency-flat gain under two matching rules. At matched BS.1770 loudness (LUFS), the proposed system’s digital A-weighted LAeq proxy is on average 1.17 dB higher than flat gain. The matched-loudness exposure hypothesis is therefore not supported. The gap is largest on bass-heavy material (about $+1.6$ to $+2.7$ dB on real bass loops). Vocals, speech, and short transients are essentially tied. The reason is straightforward: A-weighting already discards most energy near 40 Hz, so removing sub-bass lowers LUFS much more than LAeq, and flat gain must then cut the midrange to match loudness. At matched LAeq the proposed output is correspondingly quieter. All levels in this study are digital-domain proxies, not calibrated ear-level SPL.

Index Terms—Audio signal processing, A-weighting, dynamic range compression, frequency-selective limiting, loudness, multiband processing, virtual bass.

I. INTRODUCTION

Personal listening devices can deliver high playback levels for long periods. The World Health Organization and ITU-T Recommendation H.870 treat this as a cumulative A-weighted sound-dose problem and give device makers a framework for tracking exposure [1]–[4]. H.870 specifies how much dose to allow. It does not specify a signal-processing algorithm that should be used when the limit is reached.

The usual engineering response is a broadband limiter or automatic gain control [5]: when an exposure estimate exceeds a threshold, apply one scalar $G < 1$ to the whole signal. That is easy to implement and it does reduce an A-weighted metric. It is also crude. Equal-loudness contours [6], [7] show that hearing sensitivity is strongly frequency-dependent, so a flat cut changes bass, midrange, and treble by different perceptual amounts. Multiband compression and virtual-bass methods already exist as separate tools. What is missing is a direct test: if those tools are combined into an exposure-oriented limiter, do they beat “turn everything down” when exposure or loudness is matched?

That is the question of this paper. We ask:

Can frequency-selective gain allocation reduce an A-weighted digital exposure proxy more than conventional broadband attenuation at matched loudness, and preserve quality better at matched exposure?

We implement three systems on the same audio: the original clip, frequency-flat gain, and a three-band processor described in Section III. We then match either LUFS or the digital LAeq proxy and compare the other quantity. The ingredients are not new. The comparison is the contribution.

The result is mixed, and on the main hypothesis it is negative. At matched LUFS the proposed processor has a higher digital LAeq proxy than flat gain, by 1.17 dB on average over 20 clips. Bass-heavy programme accounts for almost all of the

difference. Speech, vocals, and short transients situate in the middle. Section VI shows why: the processor spends its budget on a band that A-weighting barely counts.

The rest of the paper is organised as follows. Section II reviews related methods and states the gap. Section III describes the processor as a design heuristic, not as a numerical solver. Section IV records implementation and verification. Section V defines the experiment. Section VI reports the 20-clip comparison. Section VII collects limitations in one place. Section VIII concludes.

II. RELATED WORK

A. Exposure control

H.870 defines an A-weighted sound-allowance model and discusses the difficulty of mapping a digital signal to ear-level pressure, which depends on transducer sensitivity and volume setting [1]. Device-side responses are typically volume caps, dose meters, or broadband automatic gain reduction [3], [8]. These systems manage level. They do not allocate that level across frequency in order to keep loudness.

B. Dynamic range control

Digital compressors and limiters are well understood [5], [9]. Multiband designs apply a separate envelope in each band and are usually tuned for tonal balance, broadcast loudness, or hearing-aid audibility—not for an exposure-versus-loudness comparison against a flat baseline. ITU-R BS.1770 is the standard programme-loudness meter used here as a loudness stand-in [10].

C. Loudness perception

ISO 226 equal-loudness contours [7] and excitation-pattern models [11]–[14] explain why a flat gain change is not perceptually flat. This paper uses that fact as motivation. It does not implement a Zwicker or Moore–Glasberg loudness model. Integrated LUFS is the operational loudness measure.

D. Virtual bass

When a harmonic complex loses its fundamental, listeners can still hear a pitch at f_0 [15]–[17]. Loudspeaker virtual-bass systems exploit that residue pitch by synthesising harmonics when the driver cannot reproduce sub-bass [18]–[20]. We use the same mechanism in the opposite direction: attenuate physical sub-bass and add harmonics in 80–400 Hz. Residue pitch is not the same thing as preserved bass loudness, and we do not treat it as such.

E. Gap

Table I summarises the pieces. Exposure managers, multiband compressors, and virtual-bass processors exist separately. There is little published evidence that combining them outperforms broadband attenuation when either an exposure proxy or loudness is held constant. That matched comparison is what this study provides.

TABLE I

Related methods versus this study. “Matched test” means a comparison at equal exposure or equal loudness.

Approach	Exp.	Freq.	Loud.	Bass	Match
H.870 dose limit [1]	Yes	No	No	No	No
Broadband limiter [5]	Yes	No	Ind.	No	Base
Multiband DRC [9]	Part	Yes	Part	Rare	Rare
Virtual bass [19]	No	Yes	Bass	Yes	No
This work	Proxy	Yes	Yes	Yes	Yes

III. PROPOSED METHOD

A. Design objective

Let $x[n]$ be the input. A conventional limiter applies one gain

$$y[n] = G x[n], \quad G < 1. \quad (1)$$

We want a frequency-dependent alternative that, for a given loudness, yields a lower A-weighted digital exposure proxy than (1). Preferentially cutting energy that contributes more to A-weighting than to loudness is the intended trade-off. Preferentially cutting energy that A-weighting already ignores is the failure mode.

This is a design objective, not a solved optimisation. We do not run a numerical solver for a constrained cost. The implementation is a fixed three-band heuristic built toward that trade-off.

B. Architecture

The signal is split at 300 Hz and 4 kHz, processed per band, summed, and passed through a fixed-ceiling peak limiter (Fig. 1):

$$y[n] = yL[n] + yM[n] + yH[n]. \quad (2)$$

The low band carries sub-bass, the mid band most speech and musical energy, and the high band many transients. Those cutoffs were chosen for the prototype. They were not optimised by a listening-test search.

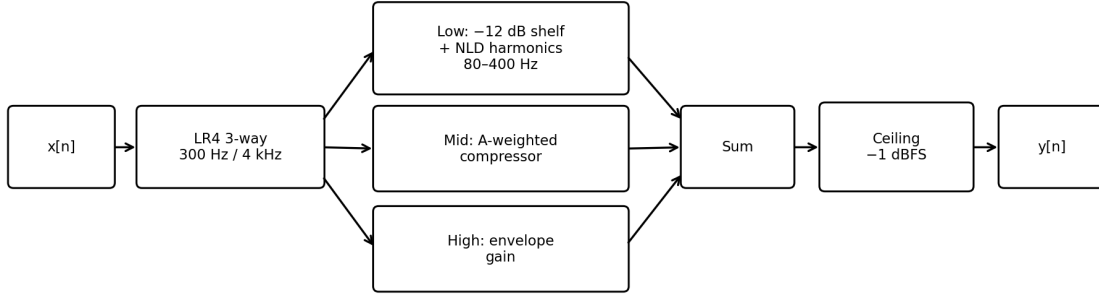


Fig. 1. Three-band processor. The crossover is complementary. Per-band processing is a fixed heuristic, not a numerical optimiser.

C. Low band

A low shelf of $\beta = 10^{(-12/20)} \approx 0.25$ (-12 dB) is applied around 100 Hz. Harmonics are generated from the isolated sub-bass x_{sub} by a memoryless nonlinearity

$$xNLD[n] = a_2 x_{sub}^2[n] + a_3 x_{sub}^3[n] \quad (3)$$

with $a_2 = 0.8$ and $a_3 = 0.35$, then band-pass filtered to 80–400 Hz and mixed back. An earlier 160–300 Hz harmonic window missed $2f_0$ and $3f_0$ for fundamentals near 40 Hz. The wider window is used so that a 40 Hz tone actually produces energy at 80 and 120 Hz (Fig. 3). Harmonic injection supports a residue pitch cue. It is not evidence that bass loudness is preserved.

D. Mid band

The mid band is compressed from an A-weighted envelope. If $e[n]$ is that envelope and

$$L[n] = 20 \log_{10}(e[n] + \epsilon) + L_{ref}, \quad (4)$$

gain reduction of ratio $R = 4$ is applied only when $L[n]$ exceeds a target of 78 (the same uncalibrated digital offset $L_{ref} = 94$ used in the exposure proxy):

$$gM[n] = 10^{[-(1-1/R) \max(L[n] - L_{tgt}, 0) / 20]}. \quad (5)$$

Attack and release are 5 ms and 50 ms. The A-weighting filter is the IEC 61672 analog prototype [21], bilinearly transformed at 48 kHz and normalised to 0 dB at 1 kHz. Inside 300 Hz–4 kHz, where this detector is used, the digital magnitude tracks the analytic curve to well under 0.2 dB. Above about 8 kHz the bilinear transform warps the response; that region is not used for mid-band detection (Fig. 4).

E. High band

Content above 4 kHz is scaled by a one-pole envelope $E[n]$:

$$yH[n] = xH[n] / (1 + \gamma \max(0, E[n] - TH)), \quad (6)$$

with $\gamma = 0.5$ and $TH = -12$ dBFS. This is transient-aware gain control. It is not a validated perceptual model.

F. Crossover reconstruction

Independent Butterworth low/band/high splits are not magnitude-complementary and can boost the sum near the cutoffs. The processor uses cascaded fourth-order Linkwitz–Riley sections [22], with allpass compensation of the low path at the upper split, so the three bands sum to an allpass. Fig. 2 compares reconstruction with the older independent-Butterworth bank: the Linkwitz–Riley sum sits at 0 dB, while the older bank peaked at the 300 Hz and 4 kHz boundaries and raised a quiet 440 Hz tone by 19.7%.

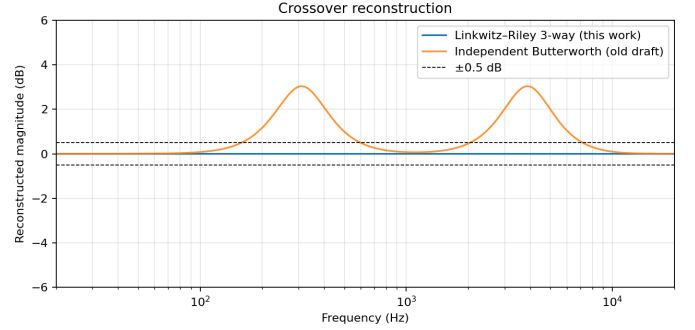


Fig. 2. Crossover reconstruction. Linkwitz–Riley 3-way sum (this work) versus independent Butterworth splits used in an earlier draft. Dashed lines mark ± 0.5 dB.

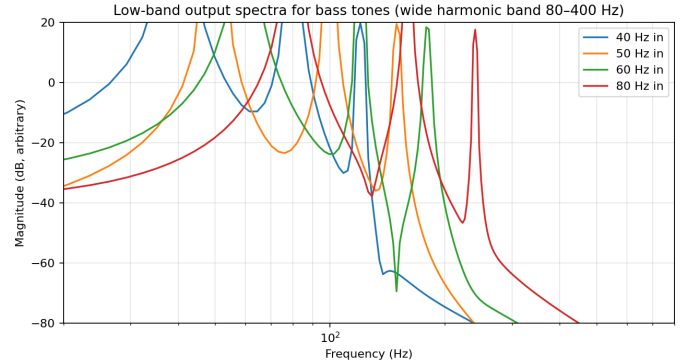


Fig. 3. Low-band output spectra for 40, 50, 60, and 80 Hz tones after the -12 dB shelf and 80–400 Hz harmonic path. Second and third harmonics are present for a 40 Hz fundamental.

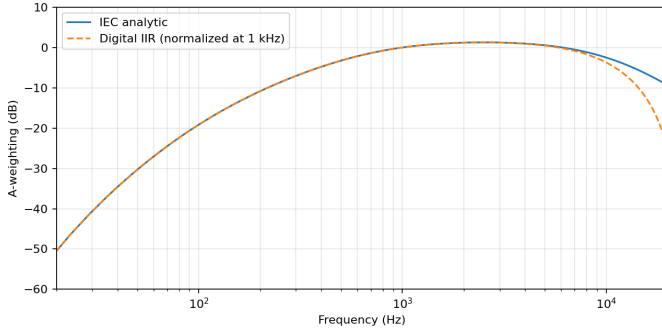


Fig. 4. Digital A-weighting IIR (normalised at 1 kHz) versus the IEC analytic curve. The mid-band detector uses 300 Hz–4 kHz. A-weighting at 40 Hz is about -35 dB, which is the mechanism behind the main result.

IV. IMPLEMENTATION

The reference system is offline Python (numpy/scipy) at $f_s = 48$ kHz. All filters are causal: second-order sections with state carried across blocks. The processor is hop-based (256 samples) and streamable. Processing a 2 s test signal in 240-sample blocks versus in one call gave a maximum absolute sample difference of 0.0. The output stage is a peak ceiling at -1 dBFS; it never renormalises a quiet block back toward full scale.

Default parameters are listed in Table II. Source code is the `exposure_dsp` package in the project repository. Listings are omitted from the paper.

TABLE II

Default processor parameters.

Parameter	Value
Sample rate	48 kHz
Crossover	LR4, 300 Hz / 4 kHz
Sub-bass shelf β	-12 dB near 100 Hz
Harmonic band	80–400 Hz ($a_2=0.8$, $a_3=0.35$)
Mid target / ratio	78 / 4:1
Mid attack / release	5 ms / 50 ms
High threshold / γ	-12 dBFS / 0.5
Output ceiling	-1 dBFS

V. EXPERIMENTAL SETUP

A. Systems

Every clip is processed by three systems:

- 1) Original: no processing.
- 2) Frequency-flat gain: one scalar G applied to the whole waveform. This is the baseline that answers “turn everything down.”
- 3) Proposed: the three-band processor of Section III at the Table II defaults.

A secondary A-weighted broadband compressor was also matched by searching its threshold. On many clips it never engaged, because the uncalibrated proxy never crossed the default target. Idle compression is not a fair “turn it down” baseline, so the reported comparison uses static flat gain.

B. Matching

Two matching rules are applied, each to within a few hundredths of a dB:

Matched exposure: G is chosen so the flat-gain output has the same digital LAeq proxy as the proposed output.

Matched loudness: G is chosen so the flat-gain output has the same integrated LUFS [10] as the proposed output (`pyloudnorm`).

Because flat gain is frequency-independent, both meters move by approximately $20 \log_{10} G$. The two matching rules are therefore two views of one difference, not two independent experiments. The interesting content is the sign of that difference and which material produces it.

C. Metrics

The digital exposure proxy is A-weighted RMS plus an arbitrary offset,

$$LAeq^{proxy} = 20 \log_{10} RMS\{wA * x\} + 94. \quad (7)$$

The offset 94 maps digital full scale to a convenient number for development. It is not a calibration to pascals or to an ear simulator. Loudness is integrated LUFS. Spectral change versus the original is log-spectral distance (LSD) over Welch power estimates; LSD is an objective distance, not a listening-test score.

D. Material

The corpus has $n = 20$ clips at 48 kHz mono: 16 Creative Commons Freesound recordings (bass loops, drum loops, choir, solo vocal, speech, cymbals) and four built-in synthetic programmes (bass-heavy, speech-like, transient, mixed). Freesound IDs: 165523, 179270, 189635, 269906, 380303, 431525, 479941, 529808, 541260, 627643, 628213, 629137, 629139, 657761, 735157, 739037. Synthetics are included for completeness and labelled as such. The conclusions below are also stated for the 16 real recordings alone.

E. Hypotheses

H2 (matched loudness): at equal LUFS, the proposed LAeq proxy is lower than flat gain.

H1 (matched exposure): at equal LAeq proxy, the proposed output is louder (higher LUFS) and/or closer to the original in LSD.

F. Listening notes

No formal MUSHRA or PEAQ test was run, and no listener scores are reported. Processed files are available as `*_proposed.wav` versus `*_gain_matched_loudness.wav` (equal LUFS) and `*_gain_matched_exposure.wav` (equal proxy). A later test should lock playback volume, randomise order, and collect ratings for overall quality, bass, clarity, and loudness. Until that is done, quality claims stay with LUFS and LSD.

VI. RESULTS

A. Matched loudness: exposure proxy

Fig. 5 plots, for each clip, the proposed digital LAeq proxy minus the flat-gain proxy at matched LUFS. Positive bars mean the proposed system is worse on the quantity it was supposed to reduce.

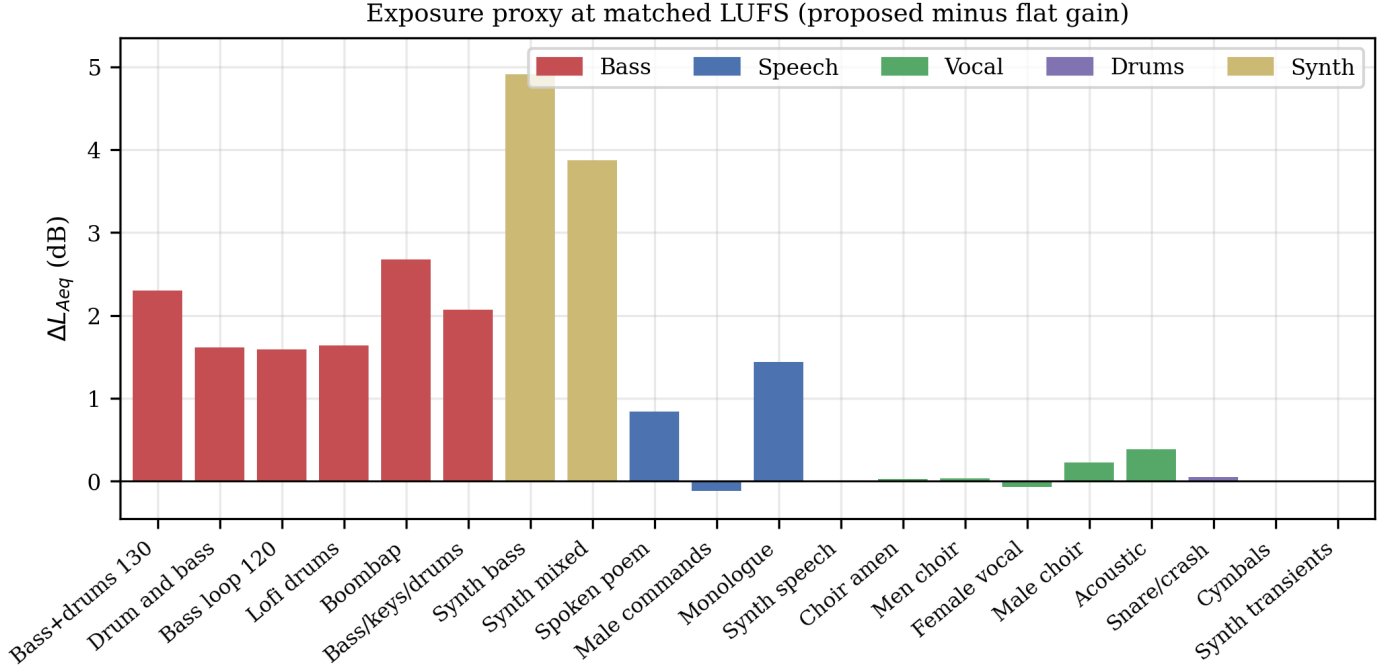


Fig. 5. Digital LAeq proxy at matched LUFS: proposed minus frequency-flat gain. Positive values mean the proposed processor has the higher exposure proxy. Bass loops (red) drive the mean; speech, vocals, and short transients sit near zero.

The mean over 20 clips is +1.17 dB (median +0.61 dB). On the 16 Freesound recordings alone the mean is +0.92 dB. Twelve clips are more than 0.05 dB worse for the proposed system, six are within 0.05 dB, and two are slightly better (a female vocal at -0.07 dB and mixed command speech at -0.12 dB). H2 is not supported.

The failure is concentrated in bass. The six real bass and drum-loop clips sit between +1.59 and +2.67 dB (mean +1.98 dB). Speech, choir, solo vocal, and cymbal clips as a group average +0.28 dB. Synthetic bass and mixed programmes are more extreme (+4.91 and +3.87 dB) because they were built with strong 40–50 Hz components; they inflate the overall mean but they do not create the finding. Real bass loops already go the same way. Per-clip numbers are in Table III.

TABLE III

Per-clip comparison. ΔLA_{eq} is proposed minus frequency-flat gain at matched LUFS (positive: proposed has higher exposure proxy). $\Delta LUFS$ is proposed minus flat gain at matched LAeq (negative: proposed is quieter). LSD is vs. the unprocessed clip at the matched-exposure condition.

Clip	Type	ΔL_{Aeq} (dB)	$\Delta LUFS$ (dB)	Murthy and Sankaranarayanan, Frequency-Adaptive Loudness Limiting	LSD prop.	LSD flat
Bass+drums 130	Bass	+2.30	-2.30		0.54	0.74
Drum and bass	Bass	+1.61	-1.61		0.70	0.19
Bass loop 120	Bass	+1.59	-1.59		0.58	1.36
Lofi drums	Bass	+1.63	-1.63		0.51	0.63
Boombap	Bass	+2.67	-2.67		0.74	0.25
Bass/keys/drums	Bass	+2.07	-2.07		0.73	0.15
Synth bass	Synth	+4.91	-4.91		0.76	0.18
Synth mixed	Synth	+3.87	-3.87		0.78	0.11
Spoken poem	Speech	+0.84	-0.85		0.49	0.20
Male commands	Speech	-0.12		+0.12	0.67	1.69
Monologue	Speech	+1.44	-1.44		0.47	0.17
Synth speech	Synth	+0.01	-0.01		0.22	0.12
Choir amen	Vocal	+0.03	-0.03		0.27	0.45
Men choir	Vocal	+0.04	-0.04		0.25	0.01
Female vocal	Vocal	-0.07		+0.07	0.57	1.40
Male choir	Vocal	+0.22	-0.22		0.37	0.08
Acoustic	Vocal	+0.38	-0.38		0.23	0.12
Snare/crash	Drums	+0.05	-0.05		0.28	0.03
Cymbals	Drums	-0.01		+0.01	0.32	0.03
Synth transients	Synth	+0.01	-0.01		0.25	0.09
Mean (n=20)		+1.17	-1.17		0.49	0.40

B. Why the proxy moves the wrong way

Fig. 6 plots, for the proposed output versus the original, LUFS drop against LAeq drop. If the two meters agreed, points would lie on the dashed line. Bass clips lie far to the right: LUFS falls by 2–5 dB while the A-weighted proxy barely moves. Speech and cymbals lie near the origin.

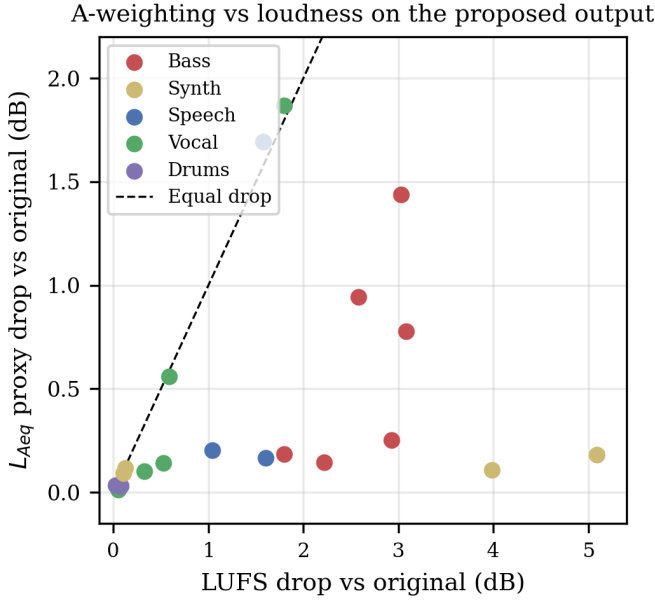


Fig. 6. Proposed processor versus the original clip. Bass-heavy material loses loudness (LUFS) without a matching drop in the A-weighted proxy, because A-weighting already ignores energy near 40 Hz.

A-weighting at 40 Hz is approximately -35 dB (Fig. 4). BS.1770 K-weighting does not discount that region to the same degree. The low-band shelf therefore does what it was designed to do—remove sub-bass—and LUFS notices. The exposure proxy does not. To match LUFS, flat gain must apply that same loudness reduction to the whole spectrum, including the midrange that A-weighting does count. Flat LAeq therefore falls further. Frequency-selective allocation aimed at “cheap” bass cuts is the wrong allocation for an A-weighted objective.

This is not a crossover bug. Reconstruction is flat (Fig. 2), harmonics are present at 40 Hz (Fig. 3), and block streaming matches whole-signal processing. The negative result is a metric mismatch, not an implementation failure.

C. Matched exposure: loudness and spectral distance

Fig. 7 is the dual view. At equal digital LAeq, the proposed output is quieter by 1.17 dB LUFS on average—exactly the opposite sign of Fig. 5, as expected for a frequency-flat baseline. H1 is not supported as a loudness-preservation claim. At the same A-weighted proxy, turning the whole signal down keeps more LUFS than cutting sub-bass.

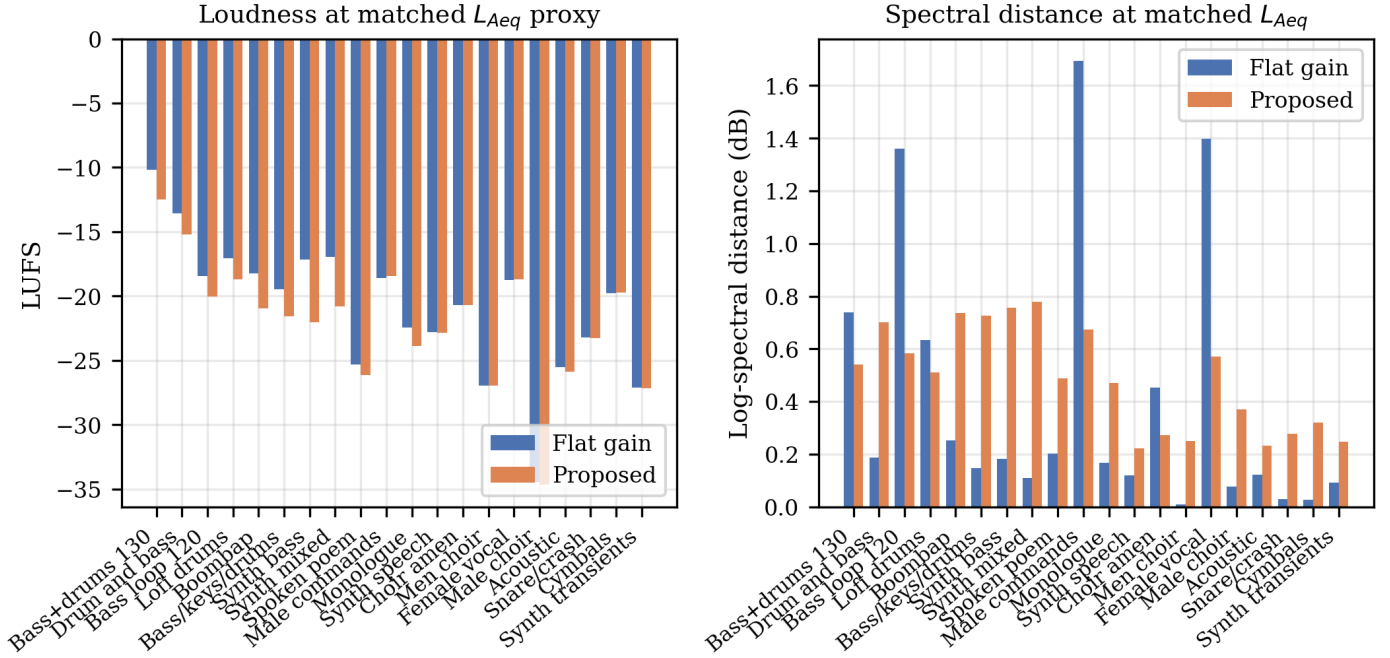


Fig. 7. Left: LUFS at matched digital L_{Aeq} proxy. The proposed system is quieter on bass-heavy clips. Right: log-spectral distance versus the original at the same match. LSD is an objective distance, not a quality grade.

Mean LSD is 0.49 dB for the proposed system and 0.40 dB for flat gain. On many bass clips flat gain barely moves, so its LSD is small. On clips where the mid-band compressor actually works (female vocal, command speech), flat gain of the same L_{Aeq} drop produces a larger spectral shift than the band-limited compressor (LSD 1.40 and 1.69 dB versus 0.57 and 0.67 dB). Spectral distance is therefore mixed: the proposed system is not uniformly closer to the original, but it avoids the largest broadband shifts. Without a listening test that pattern should not be read as “better quality.”

D. Ablation

Fig. 8 shows mean change versus the original when bands are enabled one at a time. Crossover-only reconstruction does not change level. Mid-only compression moves L_{Aeq} on speech and vocals and does little on bass. Adding the low band is what drops LUFS. Adding the high band changes almost nothing on this corpus; transients in the test set rarely exceeded the high-band threshold. The full system tracks the mid+low curve. The exposure/loudness disagreement is produced by the low shelf, which is also the block that was supposed to be the advantage.

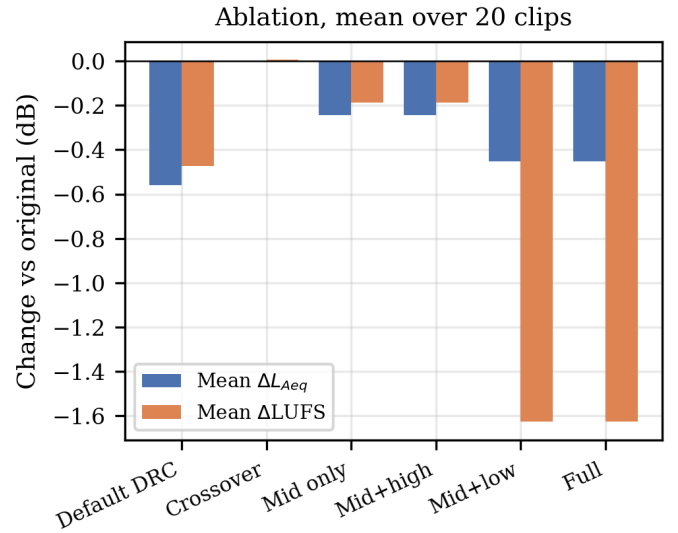


Fig. 8. Ablation, mean over 20 clips, relative to the unprocessed signal. “Default DRC” is a full-band A-weighted compressor at the default target, not the LUFS-matched flat gain of Fig. 5. The low band dominates the LUFS drop.

E. Mid-band operating point

A target/ratio sweep on a six-tone-plus-noise debug signal (Table IV) shows the expected trade-off: a lower mid target reduces the proxy and raises LSD. That sweep characterises the compressor. It is not the main result and it is not a perceptual quality measurement.

TABLE IV

Mid-band target/ratio sweep on a synthetic six-tone-plus-noise signal.

Target	Ratio	LAeq	Δ LAeq	LSD
82	2:1	77.77	-0.10	0.82
80	3:1	77.77	-0.10	0.82
78	4:1	77.04	-0.83	0.88
76	4:1	75.92	-1.96	1.16
74	6:1	74.57	-3.30	1.65
72	8:1	73.34	-4.53	2.20

VII. LIMITATIONS

This study compares two digital processing rules. It does not show prevention of hearing loss, and it does not report ear-level SPL. The LAeq figures use an uncalibrated offset; a real mapping would require a specific earphone, volume setting, and an IEC 60318-4 occluded-ear simulator [23], which still is not an individual human ear.

LUFS is a standard loudness meter, not a booth loudness match. Band edges, shelf depth, and compressor constants were not searched over listeners. No Zwicker or Moore-Glasberg model is in the loop. No PEAQ or MUSHRA scores were collected. The high band did little on this set, so the experiment mainly contrasts a low-shelf-plus-harmonics path with flat gain. Four synthetic clips are in the mean; removing them lowers the matched-loudness gap from 1.17 dB to 0.92 dB and does not reverse the sign.

VIII. CONCLUSION

We asked whether a three-band, perceptually motivated limiter could reduce a digital A-weighted exposure proxy more than turning the whole signal down, at the same LUFS. On 20 clips it did not: the proposed proxy was 1.17 dB higher on average, and 1.6 to 2.7 dB higher on real bass loops. Vocals, speech, and short transients were a tie. Cutting sub-bass spends loudness on a band that A-weighting already ignores, so a flat gain that matches LUFS wins on LAeq. The useful outcome is that comparison. If an A-weighted dose limit is the constraint, frequency-selective bass cuts are a poor allocation; any later design should spend attenuation where $wA(f)$ is large, or replace A-weighting with a metric that actually tracks the intended risk. Calibrated ear-simulator measurements and a locked-volume listening test remain the next experimental step.

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